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期末课程评教入口二维码

EE115B *Digital Circuits*

Instructor: Chenxi Xiao

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Hengzhao Yang

Zhifeng Zhu

Yajun Ha

上海科技大学

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Sequential Logic

Flip Flops

Level sensitive circuits

Level sensitive circuits

- Circuit output is controlled by clock level
- Circuit output keeps changing according to inputs when clock is active
- Circuit output value may change multiple times when clock is active

Examples

- Gated SR latch
- Gated D latch
- Clock is active high: Clk=1

Edge-triggered circuits

- Edge-triggered circuits
  - Circuit output is controlled by clock edge
  - Circuit output does not change when clock is stationary (e.g., clock remains high or low).
- Examples: Flip flops
  - D Flip Flop
  - JK Flip Flop
  - T Flip Flop

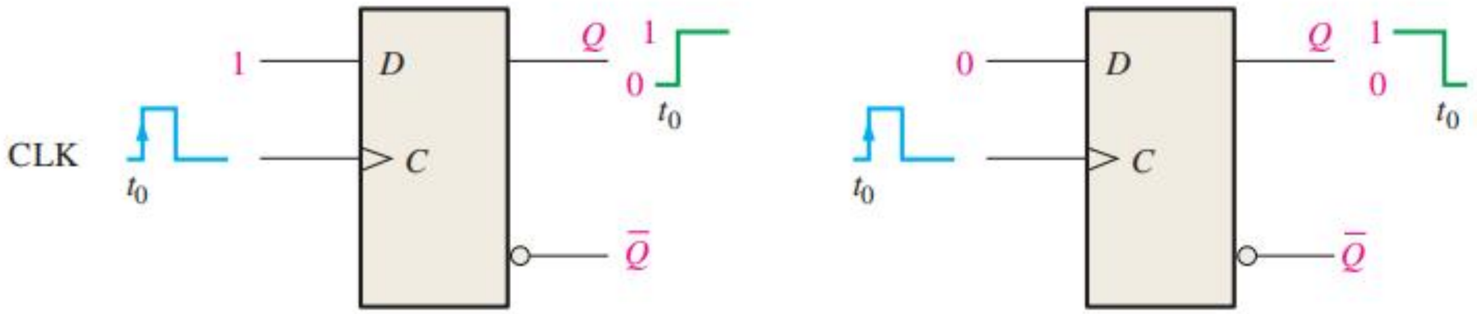
* Latch may not have a clk. It is level triggered. Flip flops are edge-triggered device with clk.

Flip-Flop

Flip-flops are **synchronous** bistable devices, also known as **bistable multivibrators.**

Flip-flops are **edge-triggered** whereas gated latches are level-sensitive. For clarity, it sometimes termed "edge-triggered flip-flops"

The D Flip-Flop



(a) $D = 1$ flip-flop SETS on positive clock edge. (If already SET, it remains SET.)

(b) $D = 0$ flip-flop RESETS on positive clock edge. (If already RESET, it remains RESET.)

FIGURE 7–14 Operation of a positive edge-triggered D flip-flop.

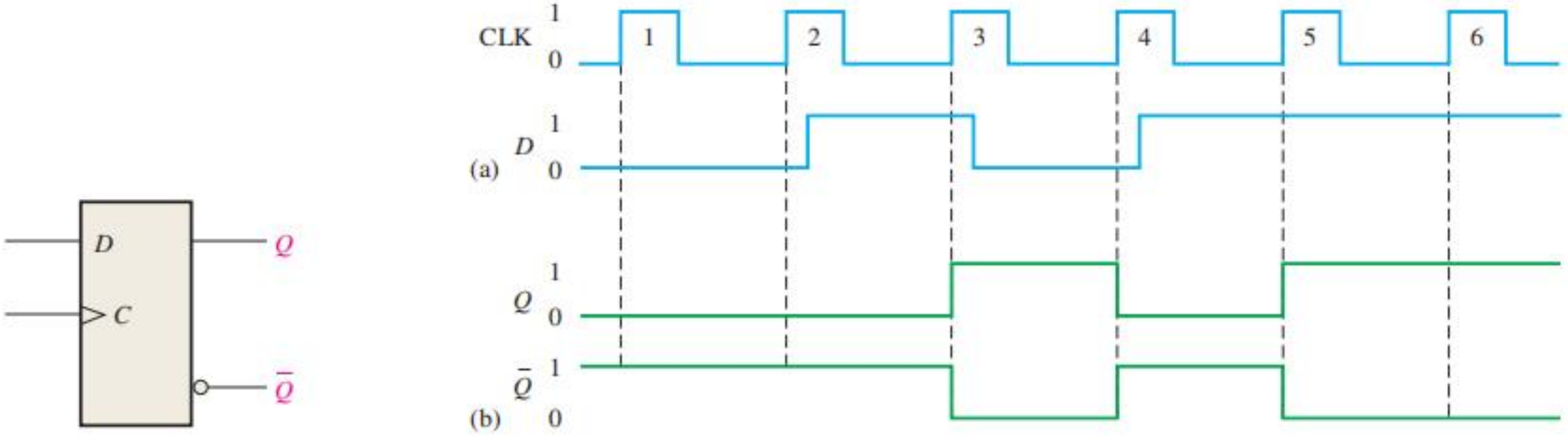
TABLE 7–2

Truth table for a positive edge-triggered D flip-flop.

Inputs		Outputs		
D	**CLK**	***Q***	$\overline{Q}$	**Comments**
0	↑	0	1	RESET
1	↑	1	0	SET

↑ = clock transition LOW to HIGH

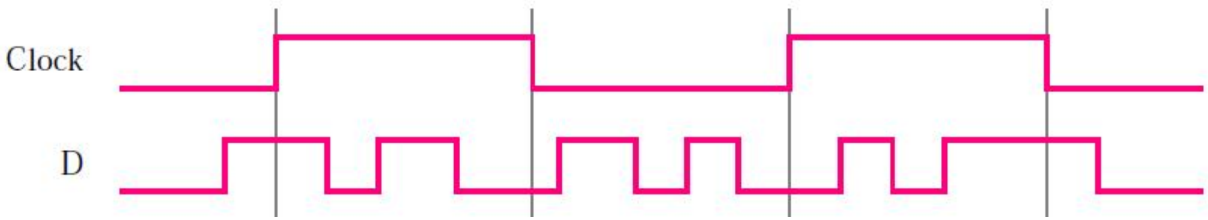
The D Flip-Flop

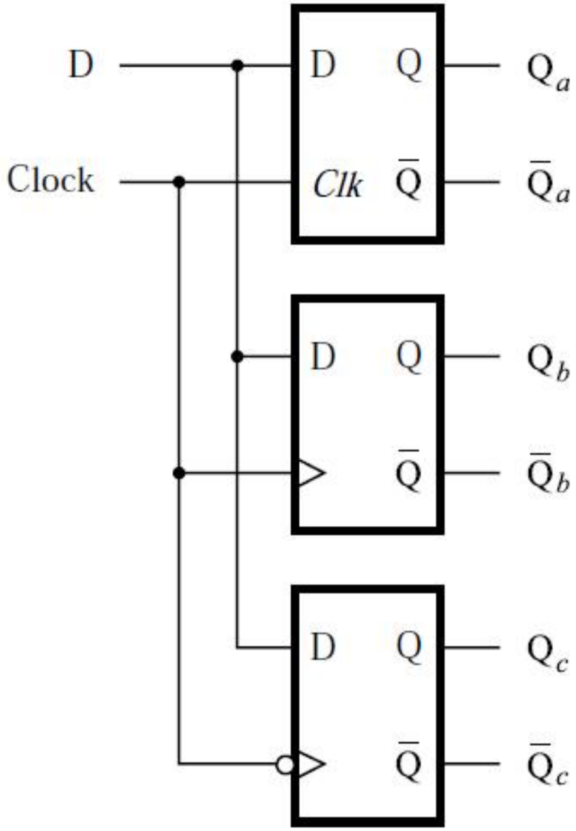


1. At clock pulse 1, *D* is LOW, so *Q* remains LOW (RESET).
2. At clock pulse 2, *D* is LOW, so *Q* remains LOW (RESET).
3. At clock pulse 3, *D* is HIGH, so *Q* goes HIGH (SET).
4. At clock pulse 4, *D* is LOW, so *Q* goes LOW (RESET).
5. At clock pulse 5, *D* is HIGH, so *Q* goes HIGH (SET).
6. At clock pulse 6, *D* is HIGH, so *Q* remains HIGH (SET).

D Latch vs. flip-flop

Exercise: Draw timing diagram of each output

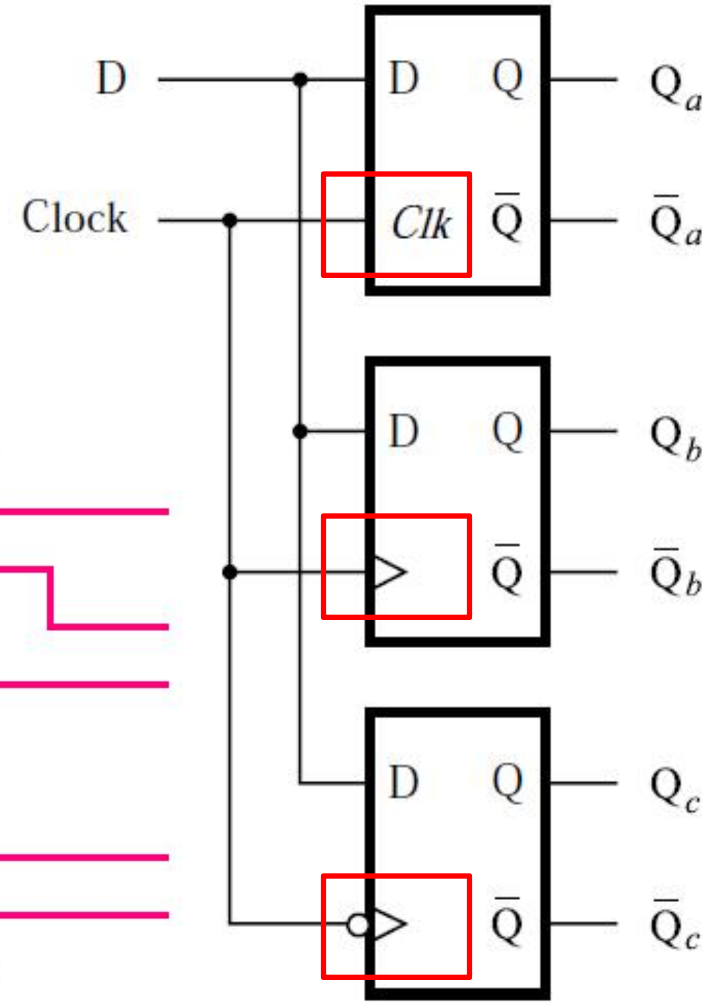


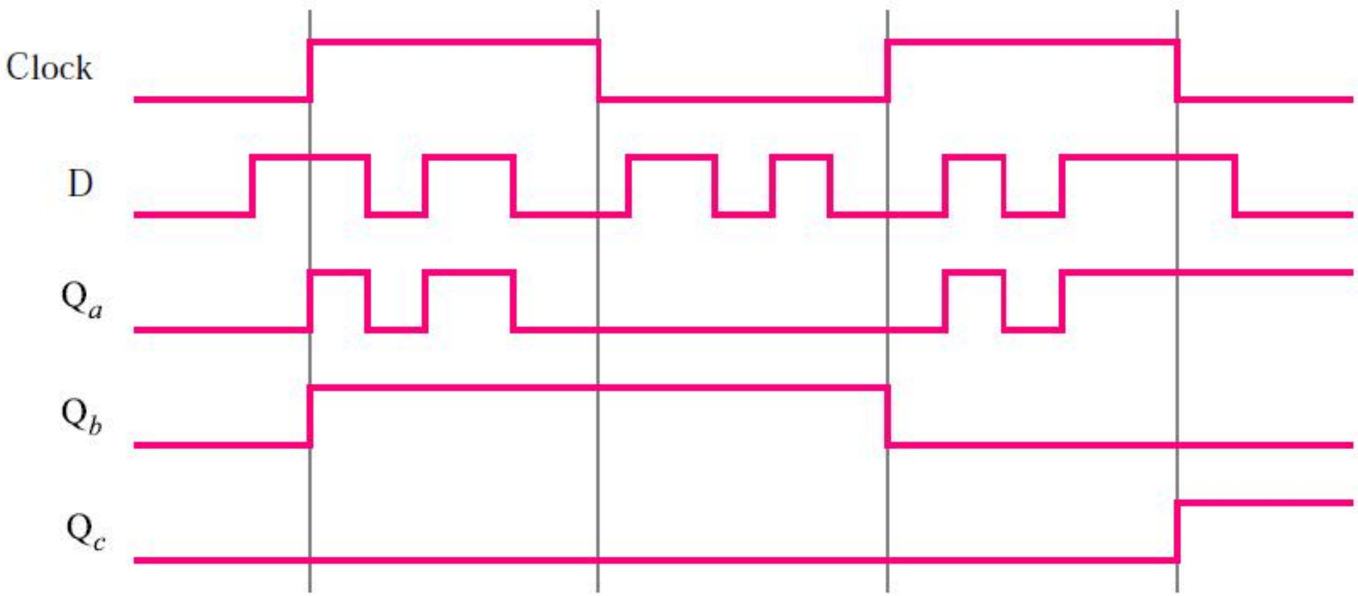


D Latch vs. flip-flop

Top to bottom

- Gated D latch (level sensitive)
- Positive-edge-triggered flip-flop
- Negative-edge-triggered flip-flop





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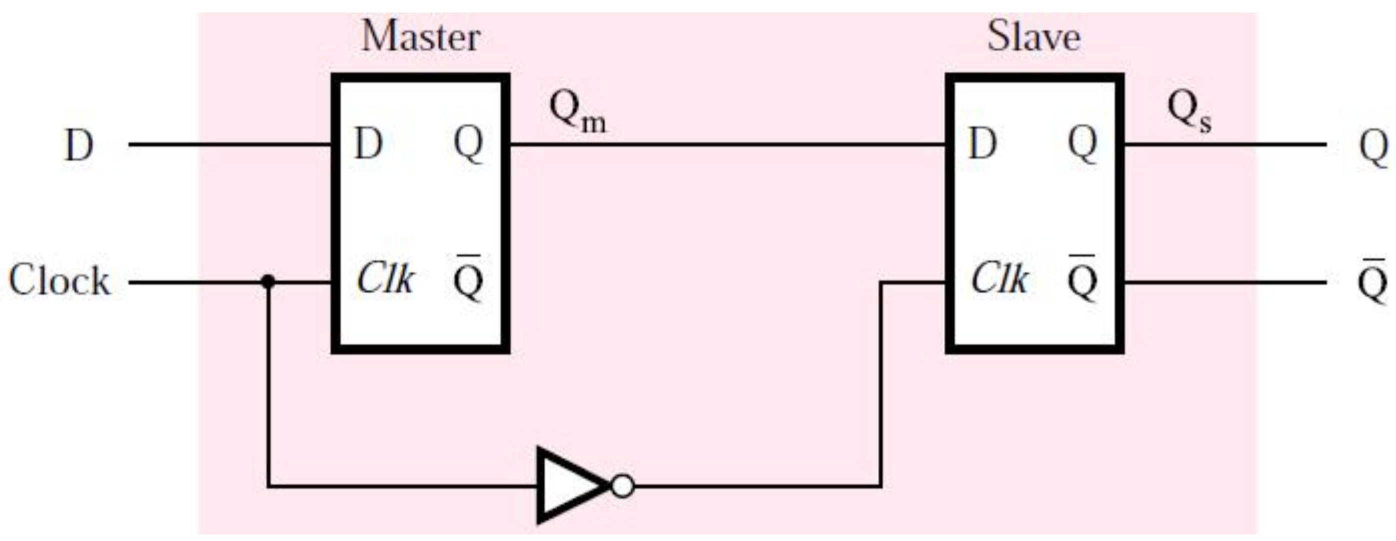
D flip-flops Implementations

- Master-slave D flip-flop
  - Built using two latches: a master latch and a slave latch connected in series.
  - Longer delay due to two stages design.
- Edge-triggered D flip-flop
  - Uses logic gates (or transmission gates) designed to respond only on the rising or falling edge of the clock.
  - Faster, but slightly more complex.

Master-slave D flip-flop: negative edge

Master-slave D flip-flop: negative edge

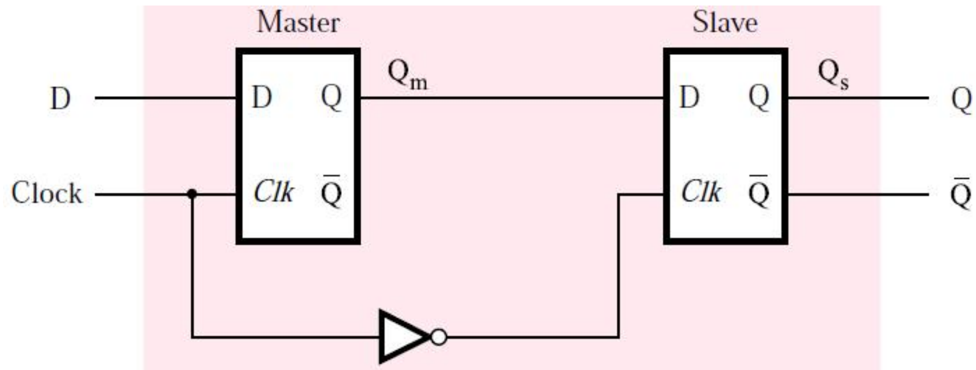
- Built on 2 gated D latches: master and slave
- Clock: original for master and complement for slave
- Output changes at negative edge of clock

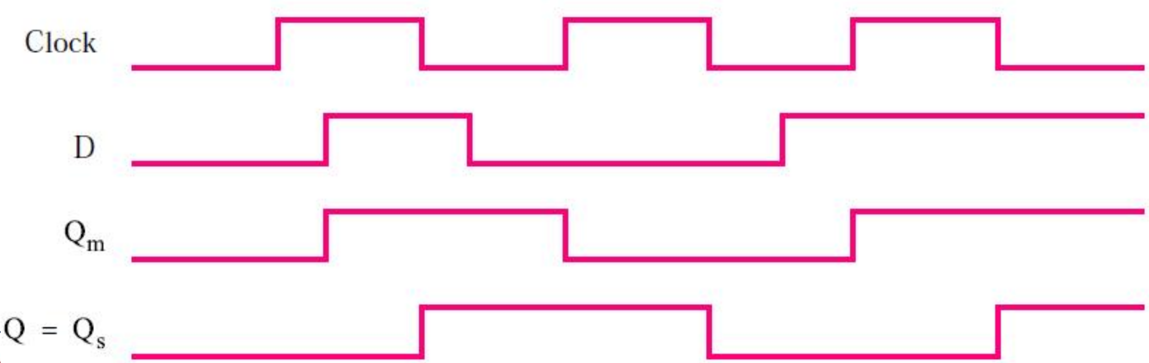


Master-slave D flip-flop: negative edge

Operation and timing diagram

- While Clock=1
  - Master is active
  - Q_m follows D
  - Q_s remains unchanged
- While Clock=0
  - Slave is active
  - Q_s responds to Q_m, Q (final output) changes at negative edge





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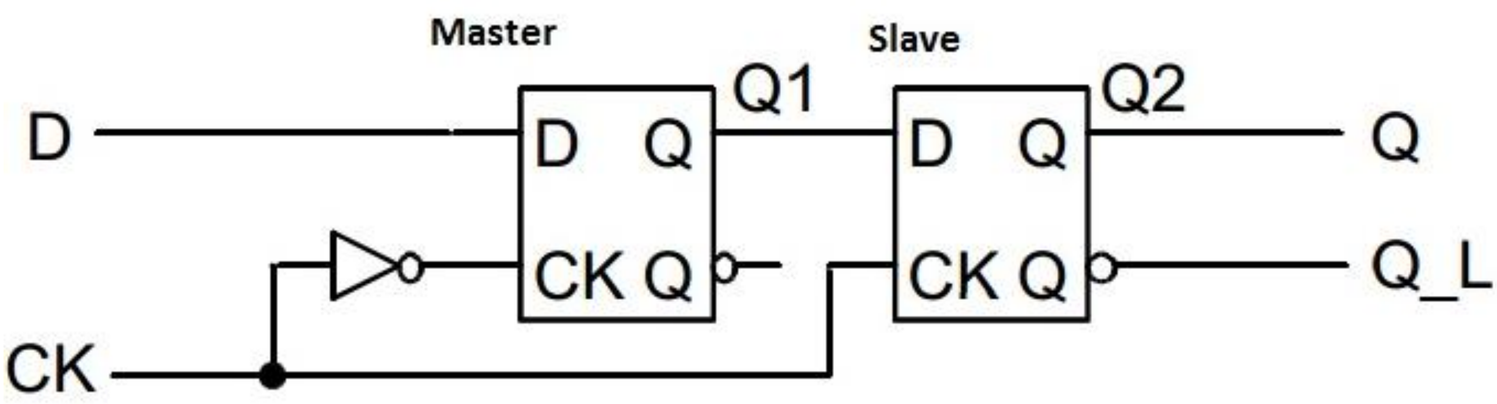
Master-slave D flip-flop: positive edge

Master-slave D flip-flop: positive edge

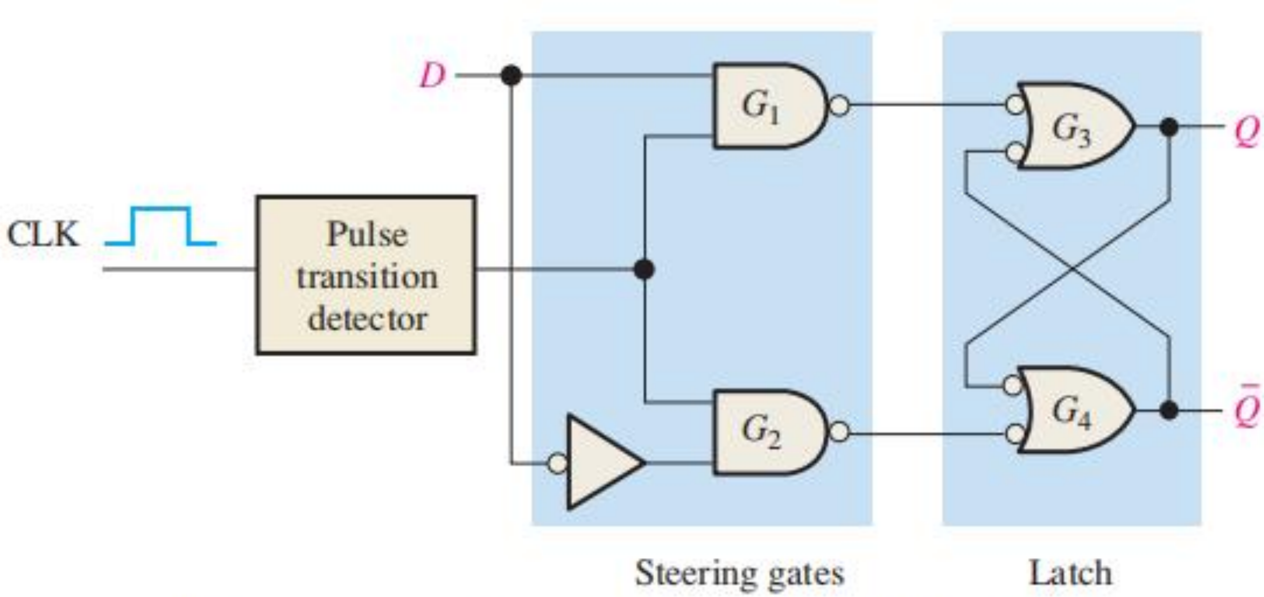
- Built on 2 gated D latches: master and slave
- Clock: complement for master and original for slave
- Output changes at positive edge of clock



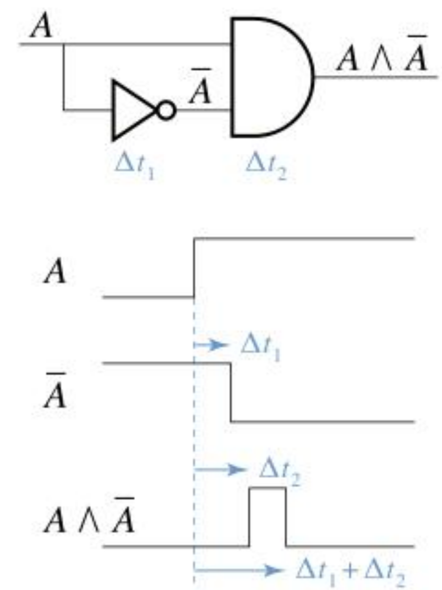
Source: https://allthingsvlsi.files.wordpress.com/2013/04/master-slave-dff.jpg

Edge-triggered D flip-flop

Enable a D Latch when pulse detected



(a) A simplified logic diagram for a positive edge-triggered D flip-flop

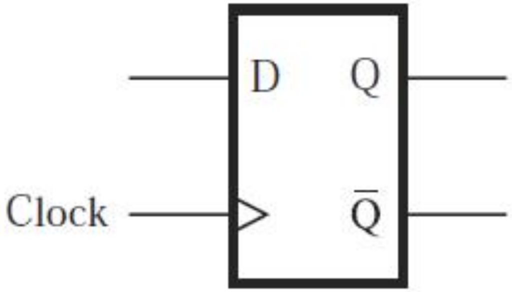


pulse transition detector

Code for D flip-flop (positive edge)

Code

- Sensitivity list: Clock
- ’EVENT: signal attribute
- Clock’EVENT: any change in the Clock signal
- Clock’EVENT AND Clock=‘1’: clock signal value has changed and the value is now 1 (positive edge)

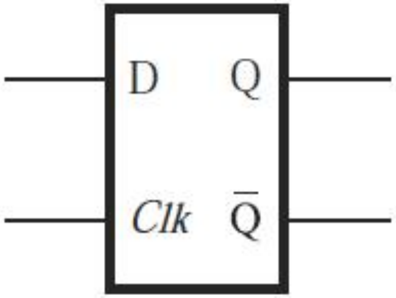



```
LIBRARY ieee ;
USE ieee.std_logic_1164.all ;

ENTITY flipflop IS
    PORT ( D, Clock  : IN    STD_LOGIC ;
           Q         : OUT   STD_LOGIC) ;
END flipflop ;

ARCHITECTURE Behavior OF flipflop IS
BEGIN
    PROCESS ( Clock )
    BEGIN
        IF Clock'EVENT AND Clock = '1' THEN
            Q <= D ;
        END IF ;
    END PROCESS ;
END Behavior ;
```

Comparison: VHDL code for gated D Latch



Clk	D	$Q(t+1)$
0	x	$Q(t)$
1	0	0
1	1	1

```
LIBRARY ieee ;
USE ieee.std_logic_1164.all ;

ENTITY latch IS
    PORT ( D, Clk  : IN    STD_LOGIC ;
           Q       : OUT   STD_LOGIC) ;
END latch ;

ARCHITECTURE Behavior OF latch IS
BEGIN
    PROCESS ( D, Clk )
    BEGIN
        IF Clk = '1' THEN
            Q <= D ;
        END IF ;
    END PROCESS ;
END Behavior ;
```

D flip-flops with CLEAR and PRESET

Sometimes we need inputs that can set or reset a flip-flop (mostly as asynchronous inputs)

- Power-On Initialization
- System reset
- Debug systems (such as when forced input is needed).

Solution: add control signals: CLEAR and PRESET

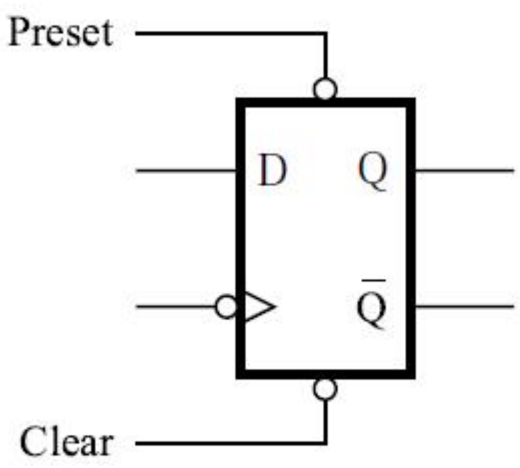
- Use CLEAR to set initial condition Q=0
- Use PRESET to set initial condition Q=1

Master-slave D flip-flop with CLEAR and PRESET

CLEAR and PRESET are active low

- CLEAR=0: Q=0
- PRESET=0: Q=1
- CLEAR=1 and PRESET=1: no effect

Circuit symbol (negative edge)



Asynchronous vs. synchronous CLEAR

Asynchronous CLEAR

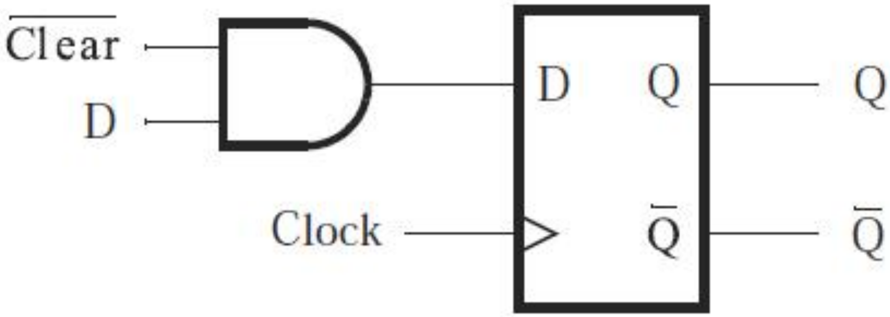
- If CLEAR=0, flip-flop will be cleared to Q=0 **immediately** without regard to clock
- May have risks of race and hazard.

Synchronous CLEAR

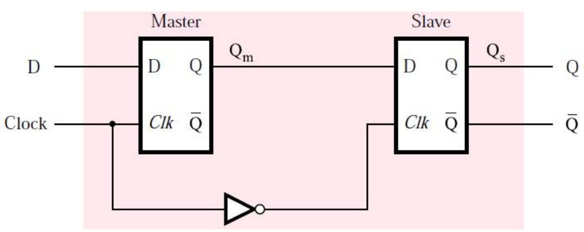
- If CLEAR=0, flip-flop will be cleared to Q=0 at the **next clock edge** (positive or negative)

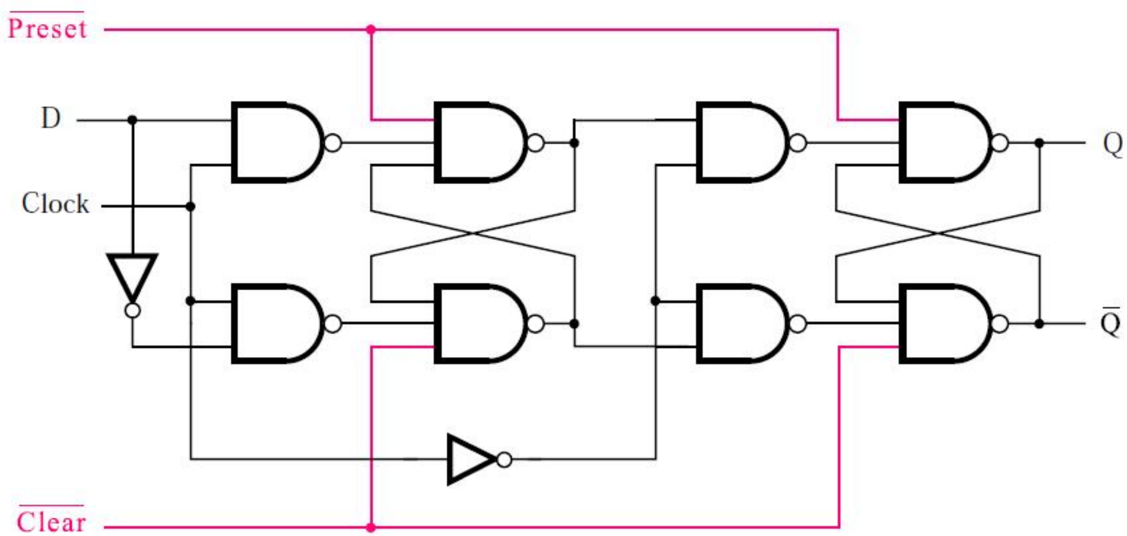
Synchronous/Asynchronous CLEAR Circuit implementation

Synchronous



Asynchronous (for master slave)





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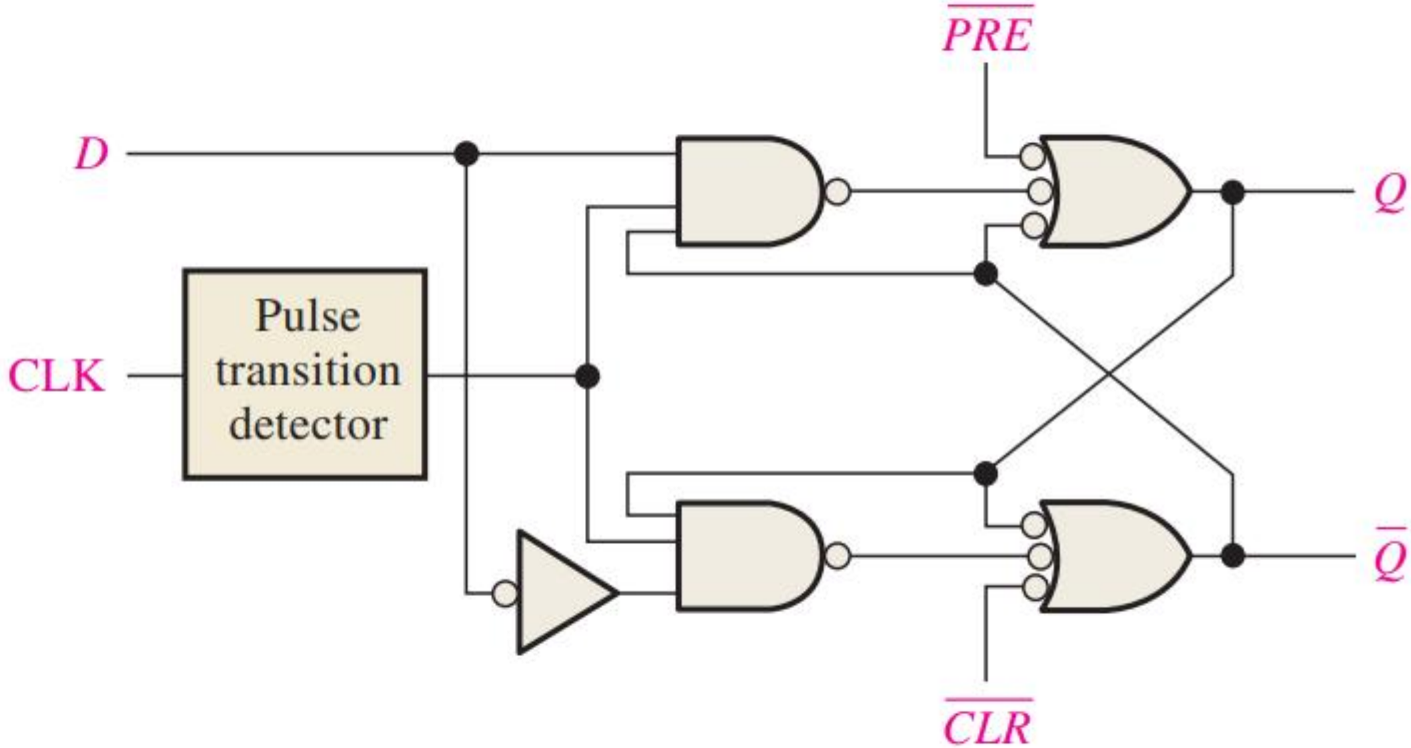
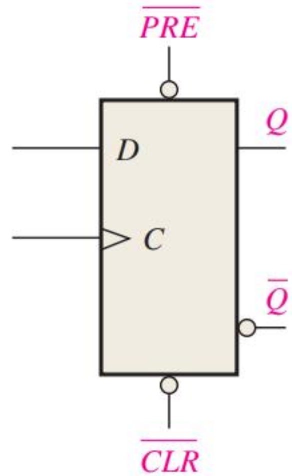
Synchronous/Asynchronous CLEAR Circuit implementation

Asynchronous (for edge-triggered)

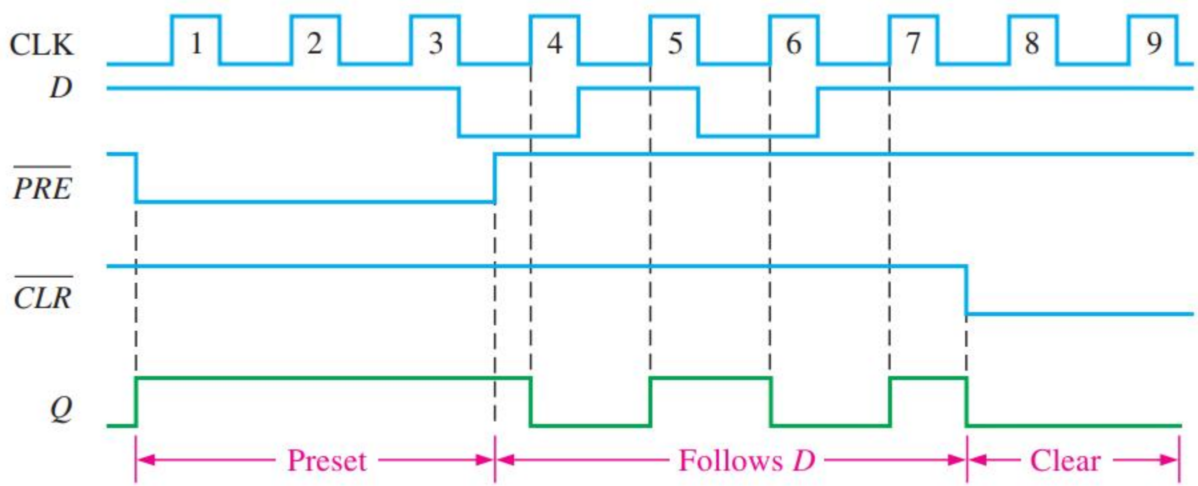


Clear/Preset: Example

Asynchronous



(a)

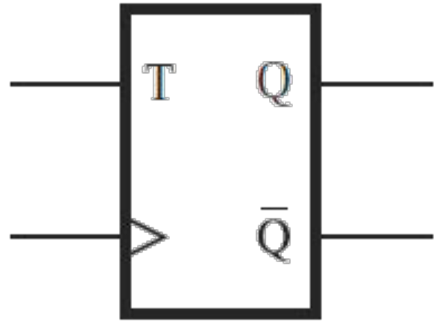


(b)

T flip-flop

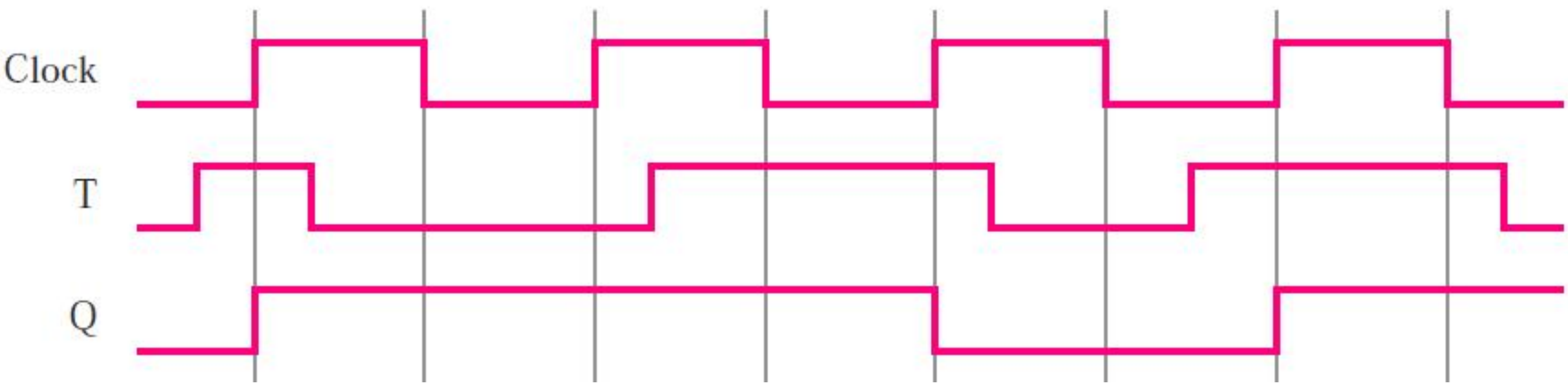
T flip-flop

- T=0: state does not change
- T=1: state is reversed



T	$Q(t+1)$
0	$Q(t)$
1	$\bar{Q}(t)$

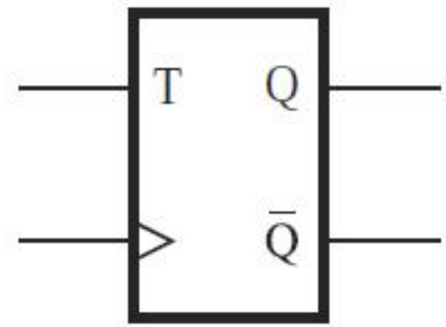
Timing Diagram



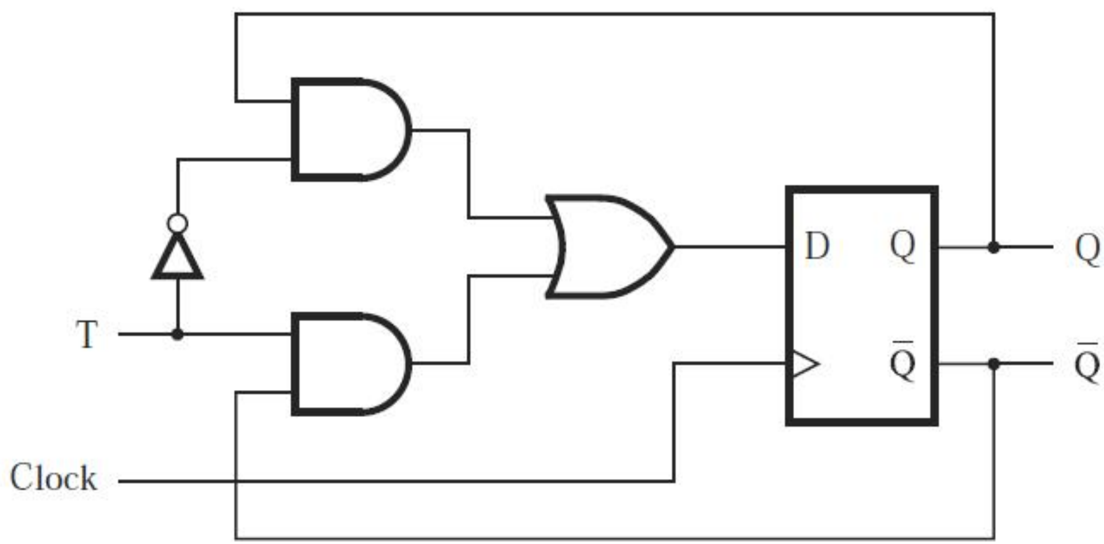
T flip-flop: Implementation

Characteristic table and circuit symbol

T	$Q(t+1)$
0	$Q(t)$
1	$\bar{Q}(t)$



Circuit (exercise: derive this using FSM)

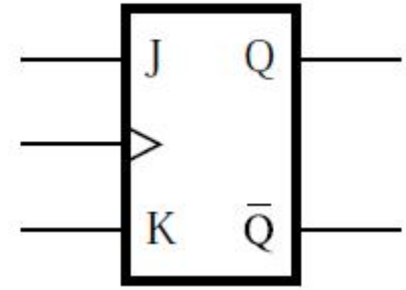


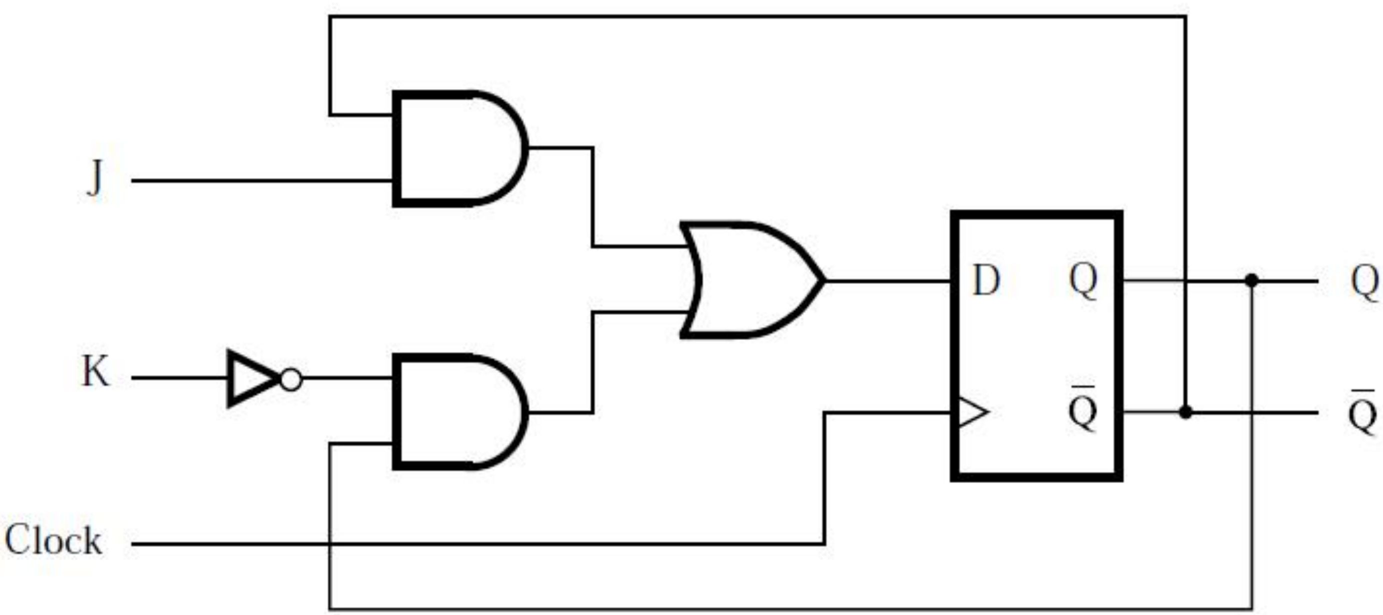
JK flip-flop

JK flip-flop

- Combines behaviors of basic SR latch and T flip-flop
- J: set, K: reset

Circuit, characteristic table, and circuit symbol





J	K	$Q(t+1)$
0	0	$Q(t)$
0	1	0
1	0	1
1	1	$\overline{Q}(t)$

(Homework: understand why this is a JK FF)